RUIJING XU

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Education

National University of Singapore **Singapore**
Master of Science, Electrical Engineering *Aug 2026 – Jun 2027*

Nanjing University of Science and Technology **Nanjing, China**
Bachelor of Engineering, Electronic and Information Engineering *Sep 2022 – Jun 2026*

Universitat Politècnica de Catalunya **Barcelona, Spain**
Exchange Student, Electrical Engineering *Sep 2024 – Jan 2025*

Internship Experience

Swancor PrimeBOT (subsidiary of AGIBOT) **Shanghai, China**
Embodied Algorithm Intern, PrimeBOT Research Institute *Feb 2026 – Jul 2026*

- **End-to-end pipeline of data, post-training and inference:** Fine-tuned Pi 0.5 and a self-developed VLA model, then deploy on the wheeled dual-arm robots Jingling G2X and Lingxi X2W.
- **Pipeline and infrastructure:** Built an automated data post-processing pipeline and error detection mechanism. Contributed to SDK and inference development for the Jingling G2X. Designed the robot zero-calibration scheme and validated cross-robot consistency.
- **Research project** (supervisor: Prof. Dong Hao @ PKU)**:** Optimizing policy generalization on long-horizon cloth-folding tasks using "SFT + DAgger" paradigm, achieving 85% success rate.

Senad Robotics Co., Ltd **Shanghai, China**
VLA Algorithm Intern, Embodied AI Research Institute *Sep 2025 – Feb 2026*

- **Pi 0 reproduction and fine-tuning:** Participated in reproducing and deploying Pi 0 on an Agile-X dual-arm. Built a data pipeline for teleoperation using Pico VR kit.
- **Algorithm development:** Led the R&D of the patent "A Cerebellar Reinforcement Learning Algorithm for High-Precision Control of Robots".
- **Embedded systems:** Completed SDK development and data visualization of a six-axis force sensor, calibration and deployment of RealSense 435i, and Linux system porting and cross-compilation.

Research Experience

Learning Visuo-Tactile Perception for Contact-rich Robotic Manipulation **Singapore**
Dissertation (Supervisor: Prof. Liu Xingyu @ NUS ECE) *Aug 2026 – Jun 2027*

- **Dataset:** Developed a UMI with visuo-tactile sensing. Constructed a contact-rich manipulation dataset covering diverse object types and materials.
- **Algorithm:** Learned visuo-tactile representations for robotic manipulation and evaluated robustness against vision-only approaches.

Edge AI-based Heart Sound Diagnosis System **Suzhou, China**
Final Year Project (Supervisor: Prof. Heng Chun Huat @ NUS ECE) *Sep 2025 – May 2026*

- **Model Design:** Designed a lightweight CNN with Coordinate Attention and a dual-Model cascade with quality-weighted averaging, achieving 92% precision.
- **Deployment:** Compressed a 65K-parameter CNN from 302 KB to 144 KB via INT8 full-integer quantization, optimized inference to 17.4 ms on ARM NEON at 1.3 % CPU utilization. The same TFLite model has a 12.8× speedup on iPhone 16 Pro.